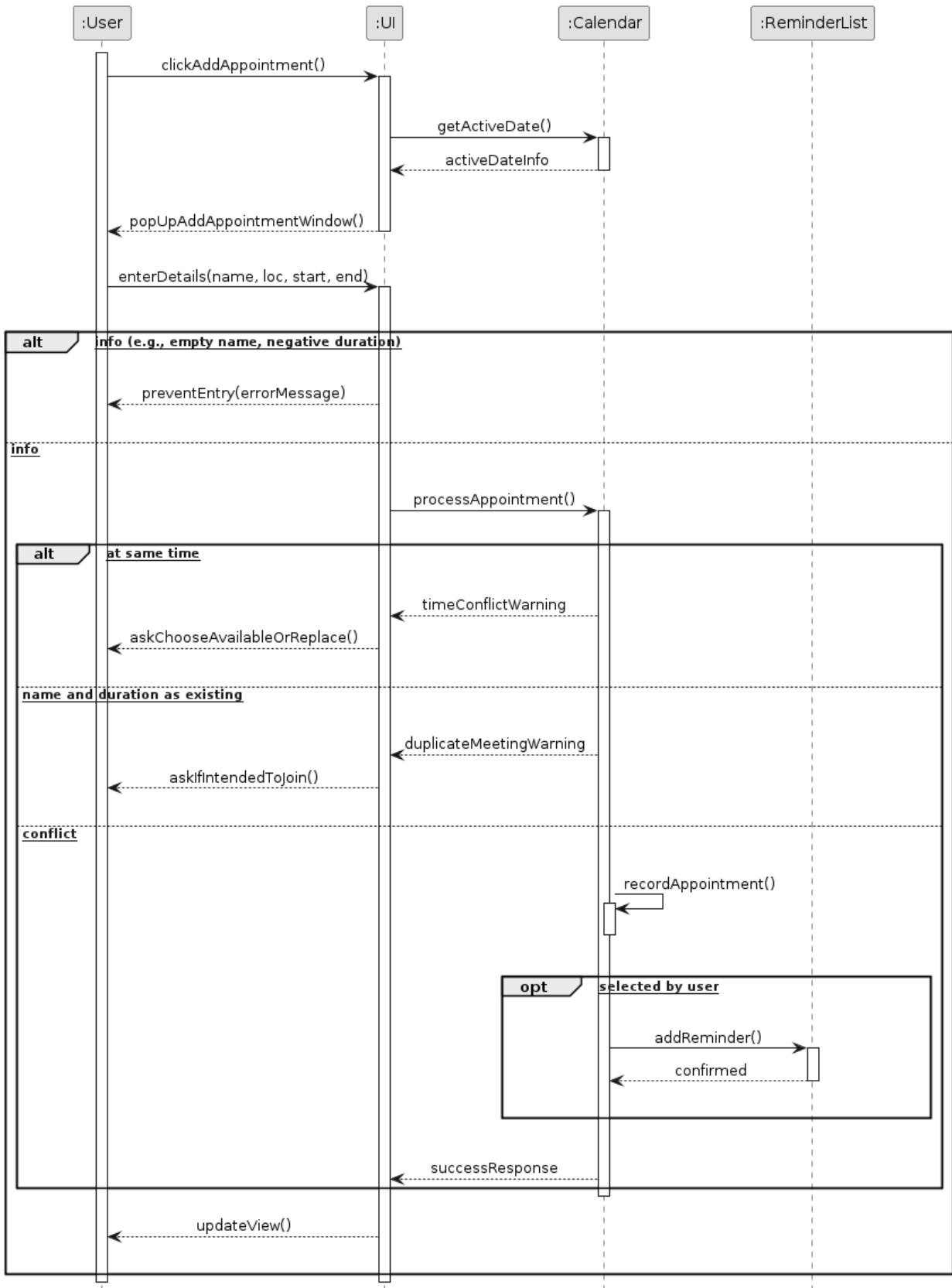


22-23

Session: 2022-2023

- **Marks:** 15
- **Question:** "Let's consider a scheduler app. The scenario begins when the user chooses to add a new appointment in the UI. The UI notices which part of the calendar is active and pops up an Add Appointment window for that date and time. The user enters information about the appointment's name, location, start and end times. The UI will prevent the user from entering an appointment that has invalid information, such as an empty name or negative duration. The calendar records the new appointment in the user's list of appointments. Any reminder selected by the user is added to the list of reminders. If the user already has an appointment at that time, the user is shown a message and asked to choose an available time or replace the appointment. If the user enters an appointment with the same name and duration as an existing meeting, the calendar asks the user whether he/she intended to join that meeting instead. If so, the user is added to that meeting's list of participants." **Draw the sequence diagram for this scenario.**

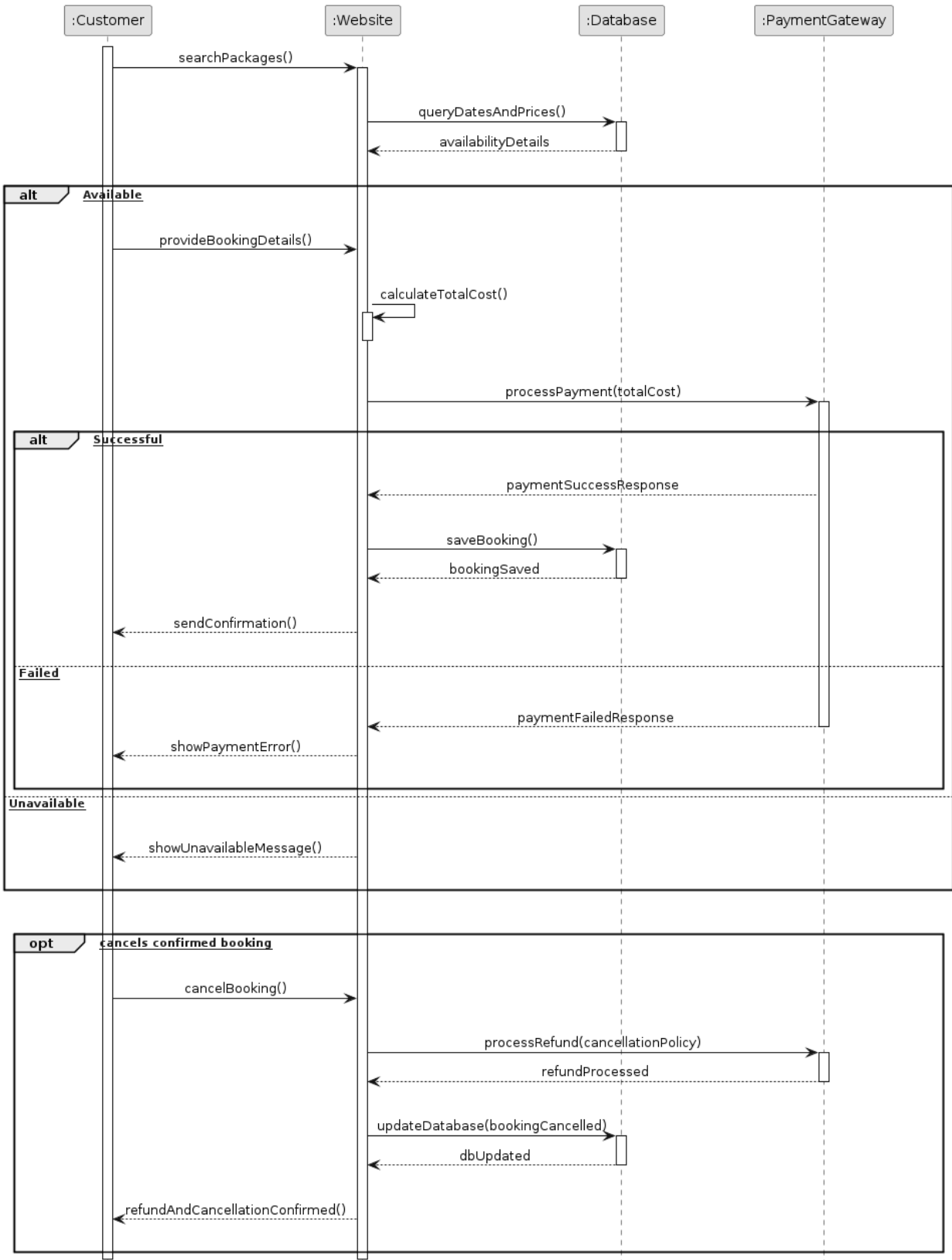


20-21

2. Session: 2020-2021

- **Marks:** 15

- **Question:** "Draw a Sequence Diagram for a Tourism Operator system. A customer searches for available tour packages on the website. The website queries the Database for available dates and prices. If the selected package is available, the customer proceeds to book it by providing details. The system calculates the total cost. The customer makes a payment via a Payment Gateway. If the payment is successful, a confirmation is sent, and the booking is saved. If the customer cancels a confirmed booking, the system processes a refund according to the cancellation policy and updates the database. Show all alternative flows (alt/opt)."

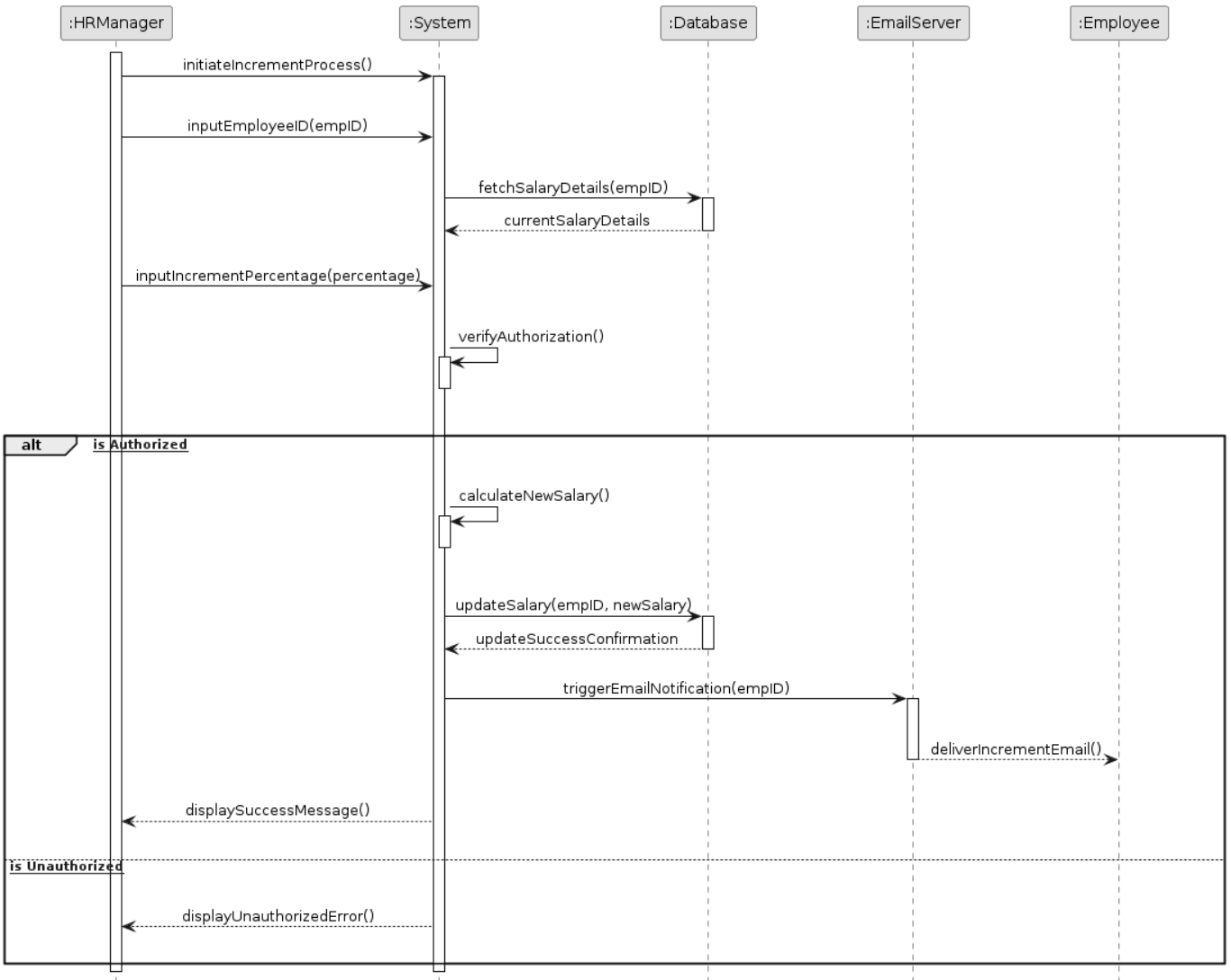


19-20

๖. Session: 2019-2020

- **Marks:** 12

- **Question:** "Consider a Payroll Management System. An HR Manager initiates the salary increment process for an employee. The Manager inputs the Employee ID. The System fetches the current salary details from the Employee Database. The Manager inputs the increment percentage. The System verifies the HR Manager's authorization. If authorized, the system calculates the new salary and updates the Database. An email notification is sent to the Employee via the Email Server. **Draw the sequence diagram.**"

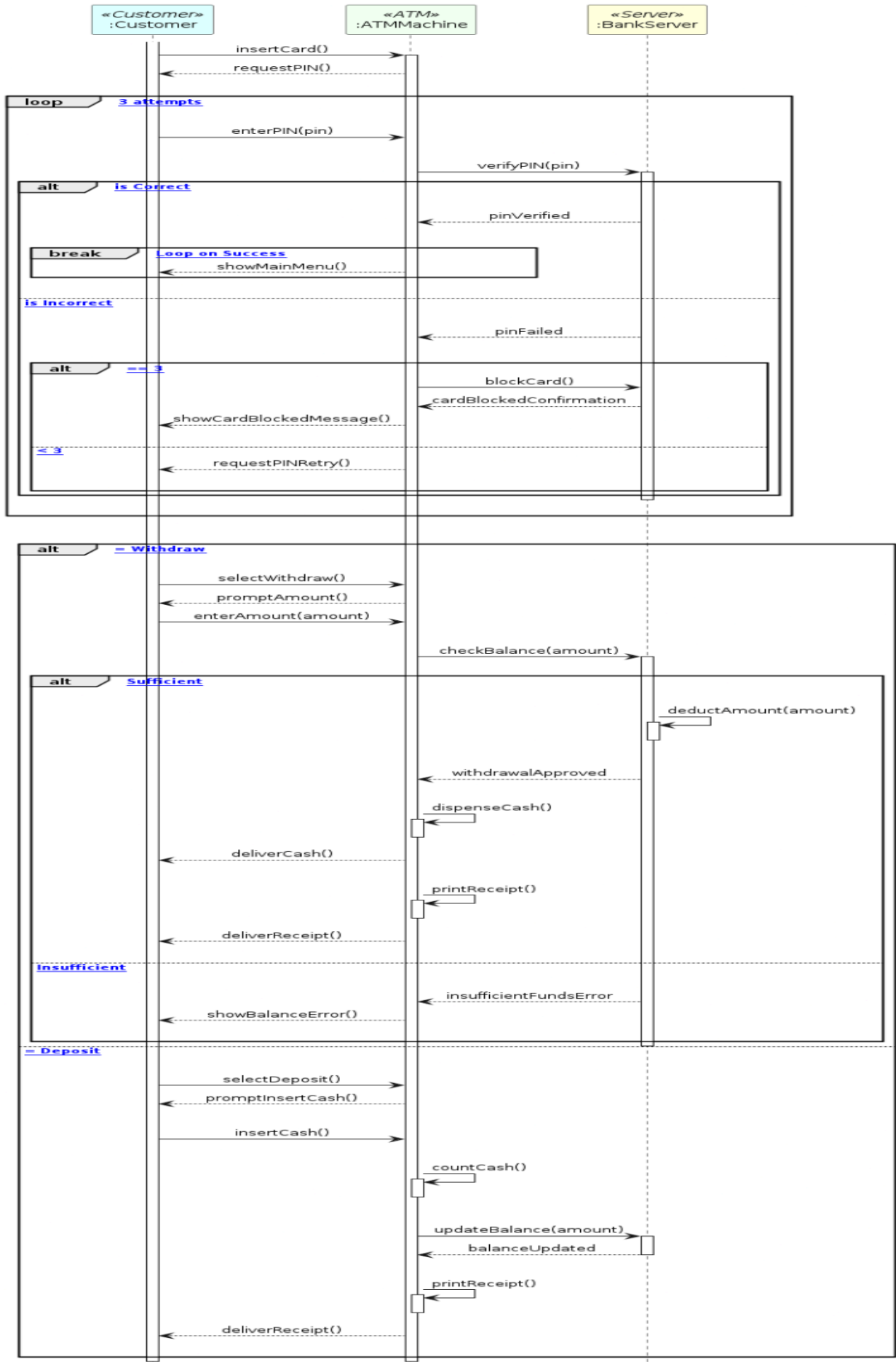


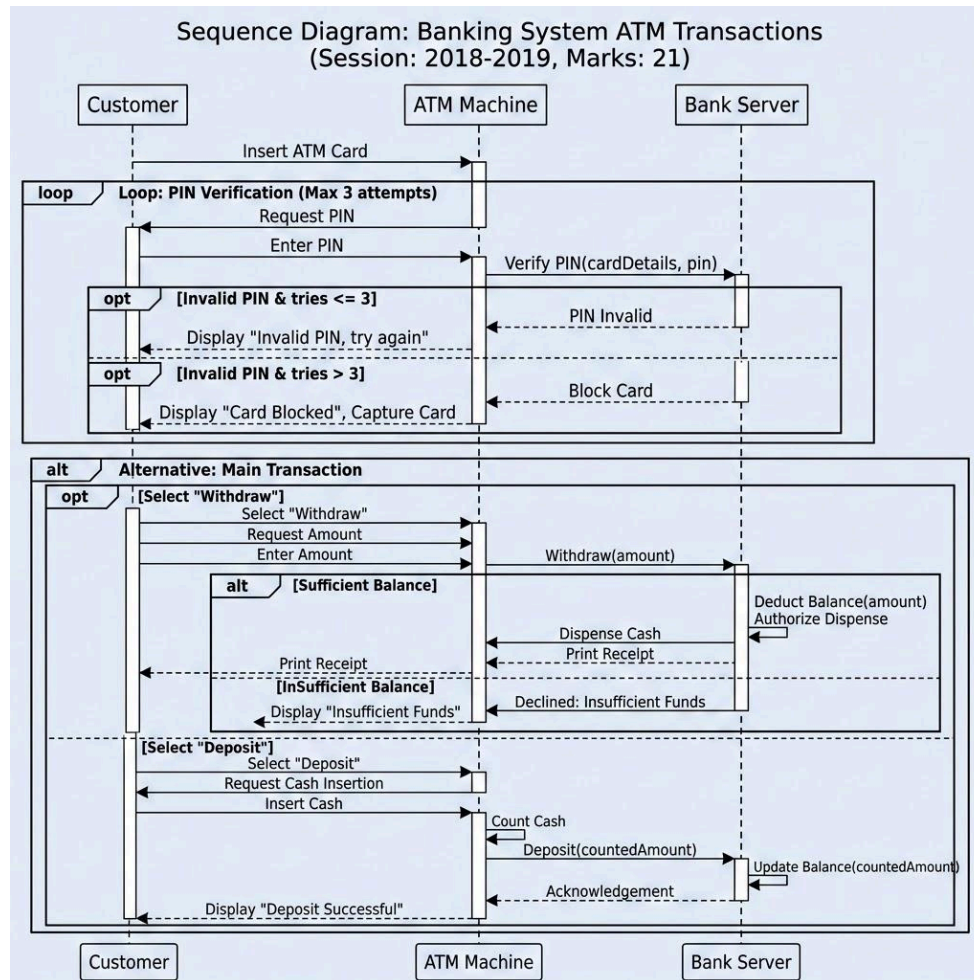
18-19

8. Session: 2018-2019

- **Marks:** 21

- **Question:** "Draw a sequence diagram for a Banking System covering the following operations: A customer inserts their ATM card into the ATM Machine. The Machine requests a PIN. The customer enters the PIN, which is verified by the Bank Server. If verification fails 3 times, the card is blocked. If successful, the customer selects "Withdraw". They enter the amount. The Bank Server checks if the balance is sufficient. If yes, it deducts the amount, dispenses cash, and prints a receipt. If the user selects "Deposit", they insert cash, the ATM counts it, and updates the balance on the server. Use loops and alt frames appropriately."





Follow the first one , not this one

17-18

Q. Session: 2017-2018

- **Marks:** 10
- **Question:** "What is the purpose of an activation box and lifeline in a sequence diagram? Explain the differences between synchronous, asynchronous, and return messages with an appropriate example using a small sequence diagram."

Sequence Diagram Components and Message Types

1. Purpose of Lifeline

In a Sequence Diagram, the vertical dashed line below an object is called a **Lifeline**.

Purpose

It represents the **existence of an object over time** during the interaction.

Time flows from **top to bottom**, indicating that the object remains alive throughout the process.

2. Purpose of Activation Box

The rectangular box drawn on a lifeline is called an **Activation Box** or **Execution Occurrence**.

Purpose

It indicates that the object is currently **active or executing a function**.

It may also represent that the object is waiting for a response from another object.

3. Differences Between Message Types

Message Type	Arrow Type	Explanation
Synchronous Message	Solid line with filled arrowhead	Sender waits for the receiver's response before continuing execution.

Asynchronous Message	Solid line with stick/open arrowhead	Sender sends the message and continues execution without waiting.
Return Message	Dashed line with stick arrowhead	Represents the response/result returned by the receiver after execution.

Examples

- **Synchronous:** Function call in programming.
- **Asynchronous:** Sending email or background notification.
- **Return:** Returning result after function execution.

